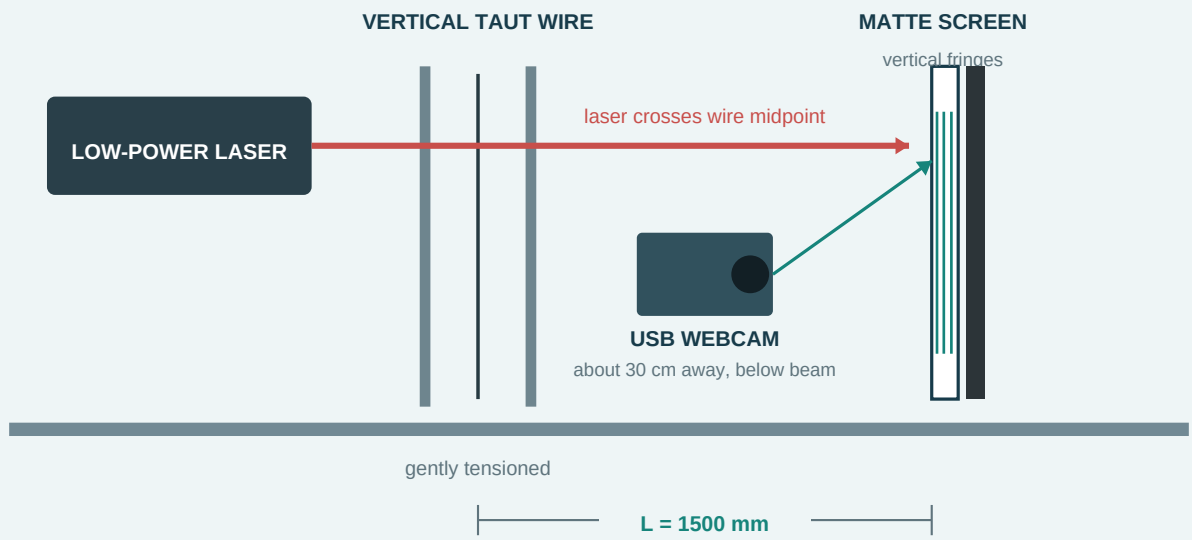
FINAL MODEL

Automatic optical wire micrometer

USB webcam + laptop analysis | Target build budget: under Rs 1,500

A laser passes a vertically held thin wire. The wire creates bright and dark diffraction bands on a matte screen. A USB webcam films the screen; laptop software finds the first dark bands and calculates the unknown diameter.

PHYSICAL BUILD - SIDE VIEW (NOT TO SCALE)



Signal path

1 Diffraction
wire creates a central bright band and dark minima

→

2 Webcam
captures the pattern on the matte card

→

3 Laptop
OpenCV extracts a brightness profile and minima

→

4 Result
diameter appears live; S saves a CSV row

No OV7670 or Pico camera bus: the laptop processes the USB webcam directly.

Use a matte card and place the camera off the beam axis. Keep a black beam stop behind the screen. Do not remove the webcam lens or aim the laser into the camera.

BUILD PLAN

Parts, budget and assembly

Make a rigid, repeatable arrangement before calibrating the software.

Parts budget

Part	Target cost	Build note
USB UVC webcam	Rs 300-500	Choose a camera that can focus on the screen at about 30 cm. Borrow/test before buying if possible.
Low-power red laser	Rs 0 borrowed	Use a labelled, teacher-approved classroom laser. Do not use an unlabelled high-power module.
Base, clamps, hood, screen	Rs 100-250	Wood/cardboard, lab clamps, matte white card, black card, and tape.
Wire/hair sample	Rs 0-50	Borrow known reference wire and screw gauge from the lab.
Laptop	Rs 0	Existing laptop runs Python and shows the result.
Expected total	**Rs 400-800**	**Shipping and optional purchases vary. Keep the project ceiling at Rs 1,500.**

Assembly - follow in this order

1. **Build the base**
   Fix the laser and wire holder to a straight wooden strip or optical bench. Keep the beam level.
2. **Hold the sample**
   Stretch the wire vertically and gently between two clamps. The laser must cross its midpoint.
3. **Set the screen distance**
   Place matte white card 1500 mm from the wire plane. Measure this distance along the beam. Put a black beam stop behind the card.
4. **Position the webcam**
   Place the webcam about 30 cm in front of the screen, slightly below the beam. Aim it at the card and focus until the pattern is sharp.
5. **Block stray light**
   Shade the screen and camera with black card. Keep the camera away from the direct beam and avoid reflective surfaces.

Pattern-size check

For a 650 nm laser, 127 um wire, and L = 1.5 m, the first-minimum pair is approximately 15.4 mm apart. If the pattern is too small in the webcam image, move the webcam closer only if it can still focus, or use the higher camera resolution.

MEASUREMENT

Calibration, calculation and presentation

The unknown sample is measured automatically after a one-time reference capture for each run.

Run the software

Install Python 3.10 or later, then install the libraries:

```
python -m pip install opencv-python numpy
```

Connect the USB webcam, close video-call apps, and start the program. Substitute the wavelength printed on the laser label; 650 nm below is an example.

```
python laser_micrometer.py --camera 0 --distance-mm 1500 --wavelength-nm 650 --reference-diameter-um 127
```

Fresh reference calibration - recommended

1. Place a reference wire with a known diameter in the holder. Its diameter should be measured with a screw gauge or supplied by a trustworthy specification.
2. Wait until the live image shows a stable central bright band and a pair of dark minima. Press **R**. The app stores the reference span for this run.
3. Replace the reference with the unknown wire without moving the laser, camera, screen, or holder. The live diameter updates automatically.
4. Press **S** to save a valid reading to `measurements.csv`; press **Q** to quit. The app asks for a fresh calibration next time it starts.

Physics relation

d = 2 lambda L / Delta-x

d = wire diameter | lambda = laser wavelength | L = wire-to-screen distance

Delta-x = separation between first dark minima on the two sides of the centre

Reference method: d_unknown = d_reference x span_reference / span_unknown

For the project demonstration

Show the live webcam image, marked minima, brightness profile, calculated diameter, and saved CSV result. Explain that the wire diffraction pattern is equivalent to the pattern from a slit of the same width (Babinet principle). Compare the reference result with the screw-gauge value.

Safety: use a labelled teacher-approved low-power laser; never look into the beam or point it at anyone.